

Root-Me Writeups

Challenge Name: Docker Layers

Difficulty: Medium

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Description: I lost the password I used to encrypt my secret flag file. Could you help me to recover it ?

Challenge Files Provided: ch29.tar

Goal: We were provided a .tar file containing a Docker image. The goal is to recover the encrypted flag and find the password used.

Methodology

1. Analyze the Docker history using container-diff

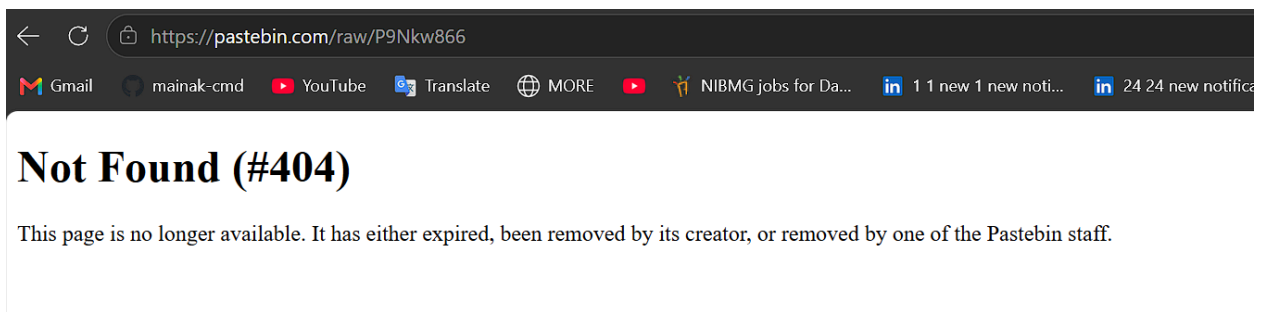
```
(sagnik@Mainak) ~/mnt/d
$ container-diff analyze -t history ch29.tar

-----History-----

Analysis for ch29.tar:
-/bin/sh -c #(nop) ADD file:d2abf27fe2e8b0b5f4da68c018560c73e11c53098329246e3e6fe176698ef941 in /
-/bin/sh -c #(nop) CMD ["bash"]
-/bin/sh -c apt update -y
-/bin/sh -c apt install -y curl openssl
-/bin/sh -c #(nop) COPY file:2ca89eb39686ffcc3d2d87bbc9293559252cfff471f80c2ed5d024b214f9a6fa3 in /
-/bin/sh -c echo -n $(curl -s https://pastebin.com/raw/P9Nkw866) | openssl enc -aes-256-cbc -iter 10 -pass pass:$(cat /pass.txt) -out flag.enc
-/bin/sh -c rm /pass.txt
```

The flag was fetched from pastebin, encrypted using aes-256 algorithm in cbc mode using a password from /pass.txt

> Visiting the pastebin : <https://pastebin.com/raw/P9Nkw866>



The flag is removed intentionally, thus we need to extract the image layers and look for **pass.txt**

2. Extract the layers

> Extract the outer tar

```
(sagnik@Mainak)-[/mnt/d]
$ tar -xf ch29.tar -C ch29

(sagnik@Mainak)-[/mnt/d]
$
```

Upon extracting the outer layer, we are introduced to the inner image layers.

```
(sagnik@Mainak)-[/mnt/d/ch29]
$ ls
1bbd61a572ad5f5e2ac0f073465d10dc1c94a71359b0adfd2c105be4c1cb2507
316bbb8c58be42c73eefeb8fc0fdc6abb99bf3d5686dd5145fc7bb2f32790229.tar
3309d6da2bd696689a815f55f18db3f173bc9b9a180e5616faf4927436cf199d.tar
4942a1abcbfa1c325b1d7ed93d3cf6020f555be706672308a4a4a6b6d631d2e7.tar
5bcc45940862d5b93517a60629b05c844df751c9187a293d982047f01615cb30
743c70a5f809c27d5c396f7ece611bc2d7c85186f9fdeb68f70986ec6e4d165f.tar
82ba49da0bd5d767f35d4ae9507d6c4552f74e10f29777a2a27c97778962476d
8d364403e7bf70d7f57e807803892edf7304760352a397983ecccb3e76ca39fa.tar
8f0d75885373613641edc42db2a0007684a0e5de14c6f854e365c61f292f3b4d
b324f85f8104bfebd1ed873e90437c0235d7a43f025a047d5695fe461da717c6.json
b58c5e8ccaba8886661ddd3b315989f5cf7839ea06bbe36547c6f49993b0d0aa.tar
ca7f60c6e2a66972abcc3147da47397d1c2edb80bddf0db8ef94770ed28c5e16
db04fe239ab708e4ab56ea0e5c1047449b7ea9e04df9db5b1b95d00c6980ff3f
manifest.json
repositories
```

> Extract the inner layers as well.

```
(sagnik@Mainak)-[/mnt/d/ch29]
$ find -type f -name "*.tar" -exec sh -c '
  for f; do
    dir="${f%.tar}"
    mkdir -p "$dir"
    tar -xf "$f" -C "$dir"
  done
' sh {} +
```

```

(sagnik@Mainak)-[/mnt/d/ch29]
$ ls
1bbd61a572ad5f5e2ac0f073465d10dc1c94a71359b0adfd2c105be4c1cb2507
316bbb8c58be42c73eefeb8fc0fdc6abb99bf3d5686dd5145fc7bb2f32790229
316bbb8c58be42c73eefeb8fc0fdc6abb99bf3d5686dd5145fc7bb2f32790229.tar
3309d6da2bd696689a815f55f18db3f173bc9b9a180e5616faf4927436cf199d
3309d6da2bd696689a815f55f18db3f173bc9b9a180e5616faf4927436cf199d.tar
4942a1abcbfa1c325b1d7ed93d3cf6020f555be706672308a4a4a6b6d631d2e7
4942a1abcbfa1c325b1d7ed93d3cf6020f555be706672308a4a4a6b6d631d2e7.tar
5bcc45940862d5b93517a60629b05c844df751c9187a293d982047f01615cb30
743c70a5f809c27d5c396f7ece611bc2d7c85186f9fdeb68f70986ec6e4d165f
743c70a5f809c27d5c396f7ece611bc2d7c85186f9fdeb68f70986ec6e4d165f.tar
82ba49da0bd5d767f35d4ae9507d6c4552f74e10f29777a2a27c97778962476d
8d364403e7bf70d7f57e807803892edf7304760352a397983ecccb3e76ca39fa
8d364403e7bf70d7f57e807803892edf7304760352a397983ecccb3e76ca39fa.tar
8f0d75885373613641edc42db2a0007684a0e5de14c6f854e365c61f292f3b4d
b324f85f8104bfebd1ed873e90437c0235d7a43f025a047d5695fe461da717c6.json
b58c5e8ccaba8886661ddd3b315989f5cf7839ea06bbe36547c6f49993b0d0aa
b58c5e8ccaba8886661ddd3b315989f5cf7839ea06bbe36547c6f49993b0d0aa.tar
ca7f60c6e2a66972abcc3147da47397d1c2edb80bddf0db8ef94770ed28c5e16
db04fe239ab708e4ab56ea0e5c1047449b7ea9e04df9db5b1b95d00c6980ff3f
manifest.json
repositories

(sagnik@Mainak)-[/mnt/d/ch29]
$

```

3. Find /pass.txt and /flag.enc

```

(sagnik@Mainak)-[/mnt/d/ch29]
$ find . -name "*pass.txt" && find . -name "*flag.enc"
./316bbb8c58be42c73eefeb8fc0fdc6abb99bf3d5686dd5145fc7bb2f32790229/pass.txt
./8d364403e7bf70d7f57e807803892edf7304760352a397983ecccb3e76ca39fa/.wh.pass.txt
./3309d6da2bd696689a815f55f18db3f173bc9b9a180e5616faf4927436cf199d/flag.enc

(sagnik@Mainak)-[/mnt/d/ch29]
$

```

4. Decrypt the flag

```

(sagnik@Mainak)-[/mnt/d/ch29]
$ openssl enc -d -aes-256-cbc -iter 10 -pass pass:$(cat ./316bbb8c58be42c73eefeb8fc0fdc6abb99bf3d5686dd5145fc7bb2f32790229/pass.txt) -in ./3309d6da2bd696689a815f55f18db3f173bc9b9a180e5616faf4927436cf199d/flag.enc -out flag.txt

(sagnik@Mainak)-[/mnt/d/ch29]
$ cat flag.txt
Well_D0ne_D0ckER_L@y3rs_Inspect0R

(sagnik@Mainak)-[/mnt/d/ch29]
$

```

Final Flag: Well_D0ne_D0ckER_L@y3rs_Inspect0R